

Paging Numericals

Calculate the size of memory if its address consists of 22 bits and the memory is 2-byte addressable.

- Number of locations possible with 22 bits = 2^{22} locations
- It is given that the size of one location = 2 bytes

Thus, Size of memory

$$= 2^{22} \times 2 \text{ bytes}$$

$$= 2^{23} \text{ bytes}$$

$$= 8\text{MB}$$

Calculate the number of bits required in the address for memory having size of 16 GB. Assume the memory is 4-byte addressable.

Solution-

Let 'n' number of bits are required. Then, Size of memory = $2^n \times 4$ bytes.

Since, the given memory has size of 16 GB, so we have-

$$2^n \times 4 \text{ bytes} = 16 \text{ GB}$$

$$2^n \times 4 = 16 \text{ G}$$

$$2^n \times 2^2 = 2^{34}$$

$$2^n = 2^{32}$$

$$\therefore n = 32 \text{ bits}$$

Consider a system with byte-addressable memory, 32 bit logical addresses, 4 kilobyte page size and page table entries of 4 bytes each. The size of the page table in the system in megabytes is _____.

Number of bits in logical address = 32 bits

Thus,

Process size

$$= 2^{32} \text{ B}$$

$$= 4 \text{ GB}$$

$$= 4 \text{ GB} / 4 \text{ KB}$$

$$= 2^{20} \text{ pages}$$

Thus,

$$\text{Number of entries in page table} = 2^{20} \text{ entries}$$

Number of Entries in Page Table-

Number of pages the process is divided

$$= \text{Process size} / \text{Page size}$$

Page Table Size-

Page table size

$$= \text{Number of entries in page table} \times \text{Page table entry size}$$

$$= 2^{20} \times 4 \text{ bytes}$$

$$= 4 \text{ MB}$$

What is the size of the physical address space in a paging system which has a page table containing 64 entries of 11 bit each (including valid and invalid bit) and a page size of 512 bytes?

without valid/invalid bit

page table entry size is 11 bits to address..page frames...11 bits means 2^{11} frames available...

no. of frames = physical address space/frame size

frame size is same as page size

*physical address space = no. of frames * frame size*

$$= 2^{11} * 512 \text{ byte} = 2^{20} \text{ B}$$

valid/invalid bit included...*10 bits are remaining...for addressing ...page frames..1bit to identify..page is valid or invalid...*

$$\text{physical address space} = 2^{10} * 512 \text{ byte} = 2^{19} \text{ byte....}$$

A computer system implements 8 kilobyte pages and a 32-bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the virtual address supported by the system is _____ bits.

8 KB pages means 13 offset bits.

For 32 bit physical address, $32 - 13 = 19$ page frame bits must be there in each PTE (Page Table Entry).

We also have 1 valid bit, 1 dirty bit and 3 permission bits.

So, total size of a PTE (Page Table Entry) = $19 + 5 = 24$ bits = 3 bytes.

Given in question, maximum page table size = 24 MB

Page table size = No. of PTEs $\times$ size of an entry

So, no. of PTEs = $24 \text{ MB} / 3 \text{ B} = 8 \text{ M}$

Virtual address supported = No. of PTEs $\times$ Page size (As we need a PTE for each page and assuming single-level paging)

= $8 \text{ M} \times 8 \text{ KB}$

= $64 \text{ GB} = 2^{36}$ Bytes

So, length of virtual address supported = 36 bits (assuming byte addressing)

Consider a machine with 64 MB physical memory and a 32 bit virtual address space. If the page size is 4 KB, what is the approximate size of the page table?

Given-

- Size of main memory = 64 MB
- Number of bits in virtual address space = 32 bits
- Page size = 4 KB

Number of Frames in Main Memory-

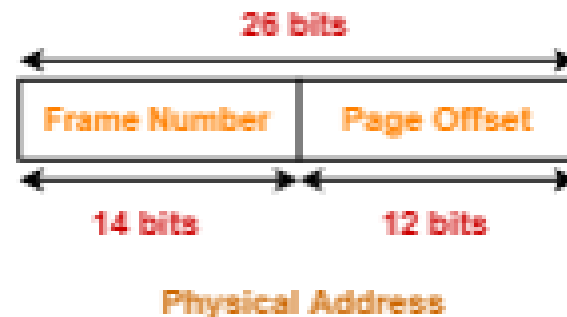
= Size of main memory / Frame size

= 64 MB / 4 KB

= 2^{26} B / 2^{12} B

= 2^{14}

Thus, Number of bits in frame number



We will consider that the memory is byte addressable.

Number of Bits in Physical Address-

Size of main memory

= 64 MB

= 2^{26} B

Thus, Number of bits in physical address = 26 bits

Number of Bits in Page Offset-

Page size

= 4 KB

= 2^{12} B

Thus, Number of bits in page offset = 12 bits

Virtual Address Space=32 Bit, Page size= 4 kb= $2^2 \times 2^{10} = 2^{12}$

PTE= V.A.S./ Page Size = 2^{20} eq. 1

Physical memory size= 64MB= $2^6 \times 2^{10} \times 2^{10} = 2^{26}$

Then No. of Frames in Physical Memory= $2^{26} / 2^{12} = 2^{14}$

And page table needs to store the address of all these 2^{14} page frames. Therefore, each page table entry will contain 14 bits address of the page frame and 1 bit for valid-invalid bit. we suppose 16 bit require it means 2 bytes (approx.)

so Size of Page table = Total PTE (from eq 1) x Size of page table entry= $2^{20} \times 2 = 2\text{MB}$

In a virtual memory system, size of virtual address is 32-bit, size of physical address is 30-bit, page size is 4 Kbyte and size of each page table entry is 32-bit. The main memory is byte addressable. Which one of the

$$\text{So, number of frames} = \frac{\text{Physical memory}}{\text{Page size}}$$

$$= \frac{2^{30}}{2^{12}} = 2^{18}$$

$$\text{So, number of bits for frames} = \log_2(2^{18}) = 18$$

We know that,

$$\text{Page table entry} = \text{Number of bits for frames} + \text{Other information}$$

$$32 = 18 + x$$

$$x = 32 - 18$$

$$x = 14$$

So, 14 bits can be used for storing protection and other information.

Assume we have a demand-paged memory. The page table is held in registers. It takes 8 ms to service a page fault if an empty page is available or the replaced page is not modified, and 20 ms if the replaced page is modified. Memory access time is 100 ns. Assume that the page to be replaced is modified 70 % of the time. What is the maximum acceptable page fault rate for an effective access time of no more than 200 ns ?

Let p = the page fault rate (the probability that a memory access results in a page fault)

Effective access time

$$= (1-p) \text{ memory access time} + \text{page fault rate} \times (\text{probability of empty or unmodified page} \\ \times \text{time to access empty or unmodified page} + \text{probability of page modification} \\ \times \text{time to access modified page})$$

We have been given that,

Effective access time = 0.20 microsecond

Memory access time = 0.1 microsecond

Probability of empty or un-modified page = 0.3

Time to access empty or unmodified page = 8000 microseconds

probability of modified page = 0.7

Time to access modified page = 20000 microseconds

Substituting these values in the formula, we get,

$$0.2 = ((1 - P) 0.1 + P(0.3 (8000) + 0.7(20000)))$$

$$\Rightarrow P = 6.1 \times 10^{-6}$$

$$= 0.00061 \%$$

Let the page fault service time be **10** milliseconds(ms) in a computer with average memory access time being **20** nanoseconds (ns). If one page fault is generated every **10^6** memory accesses, what is the effective access time for memory?

$$\begin{aligned}\text{EMAT} &= \frac{1}{10^6} \times 10 \text{ ms} + \left(1 - \frac{1}{10^6}\right) \times 20 \text{ ns} \\ &= 29.99998 \text{ ns} \\ &\approx 30 \text{ ns}\end{aligned}$$